JOEL TAVARES MARTINS

SOFTWARE ENGINEER | JS, C, JAVA

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PROFESSIONAL SUMMARY

Junior Software Engineer (3rd-year BSc student) with a strong command of **JavaScript, C, and Java**, and a special interest in **Linux systems** administration and operations. Specialized in building full-stack applications (Node.js/React) with solid foundations in algorithm development. Combines an excellent academic track record with practical experience in software development and infrastructure configuration via command-line interface (CLI). Fluent in English, I am looking to join teams where I can apply my core engineering skills to build robust and scalable solutions.

PROFESSIONAL TECHNICAL EXPERIENCE

Software & Network Engineering Intern | SPLNET, LDA — Dec 2022

- **Industrial Automation in C#:** Developed software (.NET Framework) to control a 360º garment capture system, automating camera movement and synchronizing photography.
- **Hardware Integration:** Implemented real-time control systems with serial port and IP (TCP/IP) communication, structuring and organizing data for subsequent 3D reconstruction.
- **Infrastructure & Video Surveillance:** Installed Ethernet networks and enterprise systems, including cable routing, camera configuration, NVR/DVR setup, and defined remote access via management software.

Junior Web Developer | Mixlife Digital Agency — Jan 2022 - Feb 2022

- **Full-Stack Development:** Created responsive interfaces using HTML/CSS and developed the backend using PHP, integrating with SQL databases.
- **Systems & Cloud Administration:** Configured and managed environments on Ubuntu Server and executed infrastructure operations using the AWS CLI.
- **Troubleshooting:** Actively resolved technical issues and engaged in problem-solving during solution deployment and digital asset management.

OTHER PROFESSIONAL EXPERIENCE

Front-of-House & Bar Staff | Maria Antonieta — Jun 2024 - Sep 2024

- Demonstrated reliability and teamwork in a fast-paced environment during academic breaks.
- Developed communication skills under pressure and customer service management.

Front-of-House Staff | Live Beach Mangualde (Cinderela Group) — Jul 2022 - Sep 2022

- Managed customer service and coordination in a highly dynamic, high-traffic environment.

KEY ACADEMIC & PERSONAL PROJECTS

PINT Platform & Web Architecture

- **Backend Development (Node.js/JavaScript):** Led the server-side construction for an educational web platform.
- RESTful API architecture with Express, relational database integration, and implementation of business logic and authentication.

Advanced Data Structures (Solo Project)

- **C Engineering:** Solo development of a complex data management system with native implementation of *hash tables* and *linked lists*.
- **Low-Level Control:** Dynamic memory management and allocation (malloc/free) and advanced multidimensional pointer manipulation (up to 5 levels).

Software Engineering (ES2-RAJT)

- **Collaborative Java Architecture:** Team-based development focused on applying structural and creational *Design Patterns*.
- Native implementation of the *Singleton* pattern to ensure optimized memory management and safe global instances across the application.

"EPTOKIOSK" Project (Final Year Project) 2023

🏆 **Best Project of the Year Award (Grade: 20/20)**

- Development of a comprehensive Digital Kiosk, assuming responsibility for the PHP backend creation and MySQL database modeling.

TECHNICAL SKILLS

LANGUAGES & DATABASES

JavaScript, C, Java, Node.js, React.js, C#, .NET Framework, MySQL, PostgreSQL, PHP

TOOLS & HARDWARE

Git, AWS CLI, Visual Studio, Industrial Hardware Integration, Domain Registration, Photoshop

SYSTEMS & NETWORKS

Linux Systems (Administration & CLI / Bash), Ubuntu Server, TCP/IP Suite, Serial Communication, Virtualization (VMware)

LANGUAGES

English: Fluent (C1/C2)
Portuguese: Native
Spanish: Basic (B1) | French: Basic (B1)

EDUCATION

BSc in Computer Engineering 2023 - Present
Polytechnic Institute of Viseu

Professional Technician in Computer Systems Programming & Management 2020 - 2023
Escola Profissional de Oliveira do Hospital (EPTOLIVA)